AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 24 · STUDENT LESSON

Reproducible environment and records

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Keep the materials and decisions that generated each result

Link exact inputs, script, runtime, metric and raw output without confusing replay with scientific validity.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Keep the materials and decisions that generated each result

> Link exact inputs, script, runtime, metric and raw output without confusing replay with scientific validity.

Learning goals

I can explain the **mechanism** and terminology of reproducible environment and records.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Power and resource tradeoffs. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain reproducible environment and records using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of reproducible environment and records

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Link exact inputs, script, runtime, metric and raw output without confusing replay with scientific validity.

Start with the research problem

A screenshot of a score does not tell another team which input, preprocessing, code, configuration or runtime produced it. A seed controls only the randomness that the procedure actually uses; it cannot supply missing data or dependencies. A useful run record preserves the exact materials and decisions, including failures and changed versions.

Matching a hash checks recorded bytes, not whether labels, sampling or the scientific question are correct.

Original concept explanation

A reproducible environment record links code, data, configuration and runtime dependencies to results. A seed alone is insufficient.

Complete case inputs, rules and limits

Run-record item	Complete exercise scope
Main folder	Only seed 42 and final-score.txt supplied
Missing main artifacts	Dataset and split, code, preprocessing, dependency/runtime versions, metric/settings and raw outputs
Main conclusion	Replay cannot be established from these materials
Separate mean example	Input 2, 4, 6; sum divided by count
Changed input	2, 4, 9; same formula, new input/run record
Local demonstration	Exact UTF-8 input/script bytes, SHA-256 hashes and actual Python runtime recorded in memory
Randomness	None used in the mean calculation; no seed is needed
Evidence limit	Tiny local replay is not independent AI-model reproduction or proof of valid science

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Record exact data IDs/splits, code version, dependency versions, parameters, randomness policy and raw **output**s. Verify hashes for integrity without treating them as correctness proofs.

Complete main-case evidence ledger

Main supplied artifact	What it tells us	What remains missing
Seed 42	A stated seed value	Algorithm, input, random library and how seed was used
final-score.txt	A final output record exists in the scenario	Which metric, denominator and run generated it
No dataset/split	Cannot identify evaluated cases	Exact data materials and roles
No code/preprocessing version	Cannot reconstruct transformation	Exact executable procedure
No dependencies/runtime	Cannot establish executable environment	Versions and execution instructions
Replay claim	Not established	Supply artifacts and perform a documented check

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Reproducible environment and records — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record.	Record exact data IDs/splits, code version, dependency versions, parameters, randomness policy and raw outputs. Verify hashes for integrity without treating them as correctness proofs.	Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured change

- Change: The data file changes to 2,4,9.

▼ Check after predicting

Prediction: The mean becomes 15/3=5, and the saved **input** hash must change.

Reason: The calculation is the same but its **input** records differ.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record.

Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

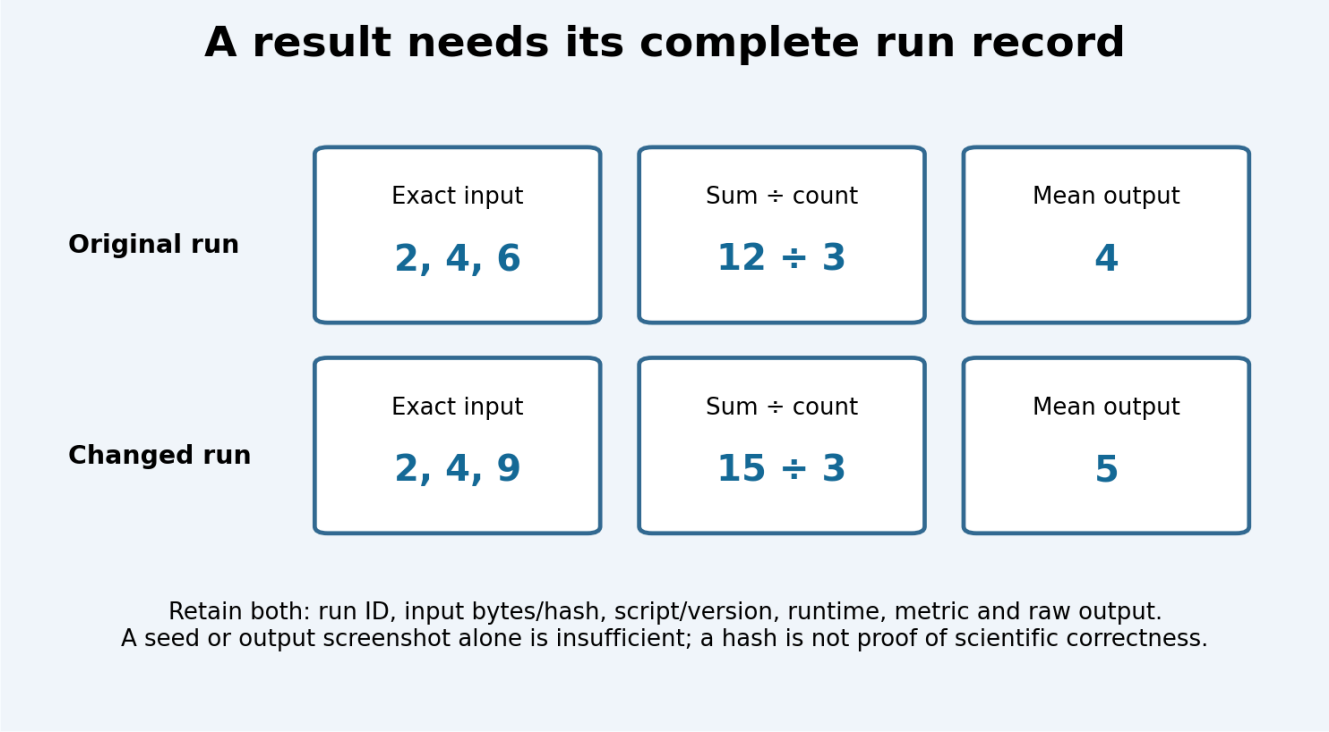
A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version.

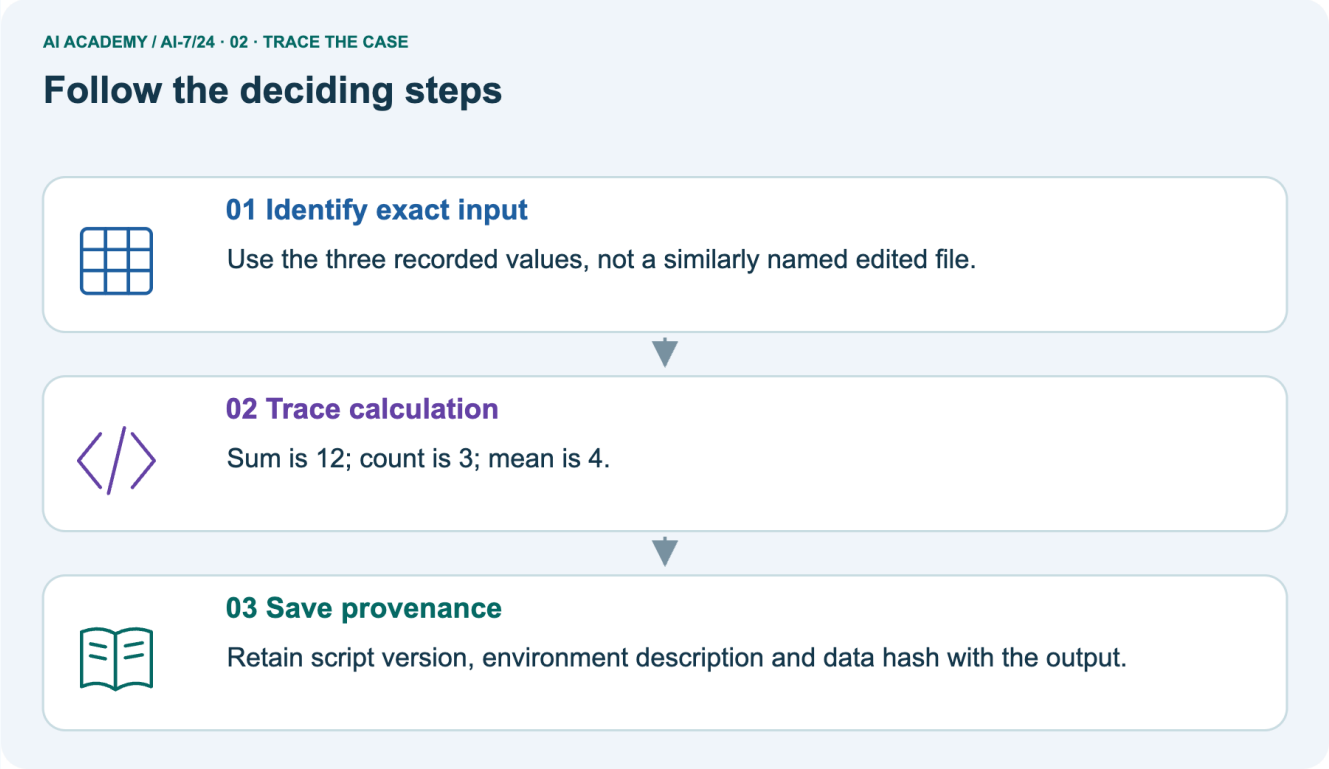
Data file contains 2,4,6. Recorded script sums values and divides by count; environment and file hash are saved.

The separate picture records a three-value arithmetic mean. For 2, 4, 6 the sum is twelve and the count is three, so the result is four. Changing the last value to nine gives sum fifteen and mean five. Retain the original record and assign a distinct run identity to the changed input. The accompanying local demonstration actually executes this tiny formula in memory and hashes its exact script and UTF-8 input bytes.

It reports the runtime it is using, not an invented environment. Matching hashes identify those bytes for this audit; they do not prove source history, correct sampling or that the formula answers the research question.

Read the labelled representation





Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify exact input	Use the three recorded values, not a similarly named edited file.
2. Trace calculation	Sum is 12; count is 3; mean is 4.
3. Save provenance	Retain script version, environment description and data hash with the output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Complete picture evidence and limits

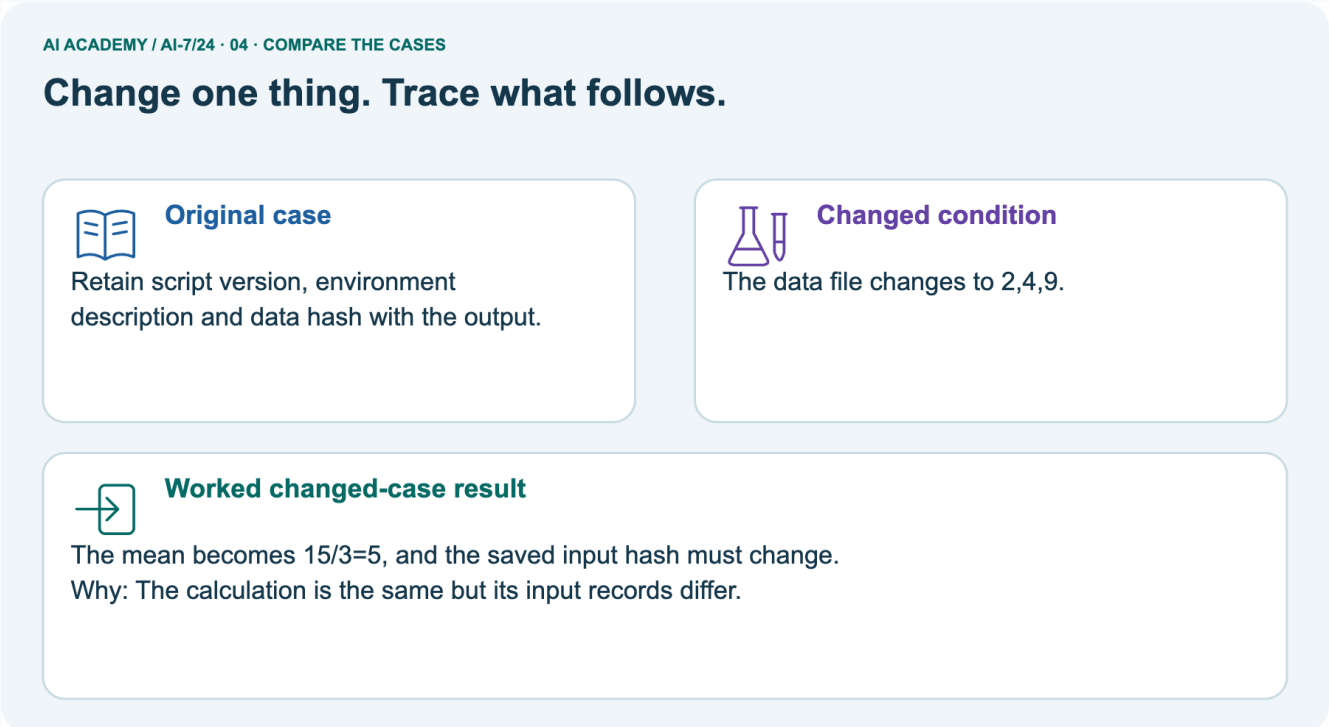
Separate mean record	Original	Changed
Input values	2, 4, 6	2, 4, 9
Sum	12	15
Count	3	3
Exact mean	12/3 = 4	15/3 = 5
Input hash	Hash of original exact bytes	Different bytes produce a different recorded hash here

Separate mean record	Original	Changed
Script hash	Same recorded formula	Same recorded formula
Run identity	Retained original	New changed-input ID
Interpretation	Local calculation only	Not proof of an AI-model reproduction

Predict the changed condition

- Change one thing: The data file changes to 2,4,9. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- Environment records support rerunning; they do not certify correct sampling, labels or scientific interpretation.

- Read the labelled picture: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.
</details>

A pictorial worked case: Reproducible environment and records

The facts we will use

AI ACADEMY / AI-7/24

Recorded mean calculation

	Value 1	Value 2	Value 3	Mean
Original data	2	4	6	4
Changed data	2	4	9	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/24

Reproducible environment and records

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify exact input

Use the three recorded values, not a similarly named edited file.





2. Trace calculation

Sum is 12; count is 3; mean is 4.





3. Save provenance

Retain script version, environment description and data hash with the output.

CHANGE ONE THING

The data file changes to 2,4,9.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

Reproducibility supports checking the computation, not automatic validity of the research claim.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Reproducibility audit	Checked interpretation
Seed and final score only	Missing materials prevent establishing replay
Same seed, different preprocessing	Not the same recorded procedure
Data content changes	Retain original and record changed input/version/run
Hash matches	Checks recorded bytes, not scientific correctness or source history
Same output from two runs	Different inputs can share outputs; output alone does not identify a run
Environment list provided	Still needs materials and execution evidence
Tiny mean re-executed locally	A demonstrated local replay, not an independent full-study reproduction
New run overwrites old ID	Breaks traceability; preserve both versions and lineage

Write and justify the repair

- A report omits this qualification: Reproducibility supports checking the computation, not automatic validity of the research claim.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the mechanism with progressively less help

> Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported workbook trace

Use workbook W2 for the connected example “Reproducible environment and records”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Reproducible environment and records. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?

> **My explanation using a pictured step and the given facts**

Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record.
Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. First audit the seed-and-score folder on paper: name each missing artifact and state that replay is not established. Then predict the two supplied means before running or tracing the complete local demonstration. It executes its recorded formula on exact in-memory text, computes input and script hashes, and captures the actual Python runtime.

   The original and changed records remain separate, with outputs four and five. This performs no network requests or external file collection, and it is not a replay of the missing AI experiment. A hash does not validate the method or authorize access to unavailable materials. Preserve the independent preprocessing/version scenario for its first attempt.

```
"""Replay a tiny self-contained calculation in memory; no external experiment."""
import hashlib
import sys
from fractions import Fraction
SCRIPT='''from fractions import Fraction

def mean_csv(text):
    parts=text.strip().split(',')
    if not parts or any(not p.strip() for p in parts):
        raise ValueError('nonempty comma-separated integer values required')
    values=[int(p) for p in parts]
    return Fraction(sum(values),len(values))
'''
namespace={};exec(SCRIPT,namespace)
def record(run_id,data_text):
 if not isinstance(run_id,str) or not run_id:raise ValueError('run ID required')
 if not isinstance(data_text,str):raise ValueError('explicit input bytes required')
 data_bytes=data_text.encode('utf-8');script_bytes=SCRIPT.encode('utf-8')
 return {'run_id':run_id,'data_text':data_text,'script_text':SCRIPT,
         'data_sha256':hashlib.sha256(data_bytes).hexdigest(),
```

```
          'script_sha256':hashlib.sha256(script_bytes).hexdigest(),
          'python_runtime':sys.version,'dependencies':'Python standard library only',
          'metric':'exact arithmetic mean of parsed integers','randomness':'none used',
          'raw_output':str(namespace['mean_csv'](data_text))}
original=record('demo-original','2,4,6\n');changed=record('demo-changed','2,4,9\n')
assert original['raw_output']=='4' and changed['raw_output']=='5'
assert original['data_sha256']!=changed['data_sha256'] and
original['script_sha256']==changed['script_sha256']
assert record('demo-replay',original['data_text'])['raw_output']==original['raw_output']
print('Original local in-memory run record:',original)
print('Changed local in-memory run record:',changed)
print('Changed input gets a different run ID; original record retained. Hashes describe
these exact bytes only.')
print('This is an actual tiny local calculation, not an independent reproduction of an AI
model, evidence of good sampling or proof that the recorded formula answers a scientific
question.')
```

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try the selected new case independently

> Explain which original and changed artifacts, versions and run identities must be retained when preprocessing and data both change.

Complete independent task

Partner changes pre**process**ing and data version while keeping old run ID. What must be preserved?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply a new case independently

Attempt this without hints

Partner changes pre**process**ing and data version while keeping old run ID. What must be preserved?

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

</details>

MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

Add a labelled reproducible environment and records **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reproducible environment and records.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of reproducible environment and records

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Environment manifest	Runtime and dependency records
Artifact hash	A digest for comparing exact content
Run identity	A versioned link between settings and outputs

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Reproducibility supports checking the computation, not automatic validity of the research claim.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Reproducible environment and records — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record.	Record exact data IDs/splits, code version, dependency versions, parameters, randomness policy and raw outputs. Verify hashes for integrity without treating them as correctness proofs.	Replay cannot be established. Supply artifacts/versions and exact metric/settings; seed 42 alone cannot reconstruct the experiment.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version.

By the end, I can	How I will show it
Explain the mechanism and terminology of reproducible environment and records	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version.
- Which labelled step matches this detail? Retain script version, environment description and data hash with the **output**.
- 1. Trace calculation
- 2. Identify exact **input**
- 3. Save provenance
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Save provenance Retain script version, environment description and data hash with the **output**.

Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Changed-case question and optional hint

- Changed-case practice: The data file changes to 2,4,9. Which conclusion is supported by the supplied facts?
- 1. The mean becomes 15/3=5, and the saved **input** hash must change.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The mean becomes 15/3=5, and the saved **input** hash must change. The calculation is the same but its **input** records differ.

Repair and retain the record

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

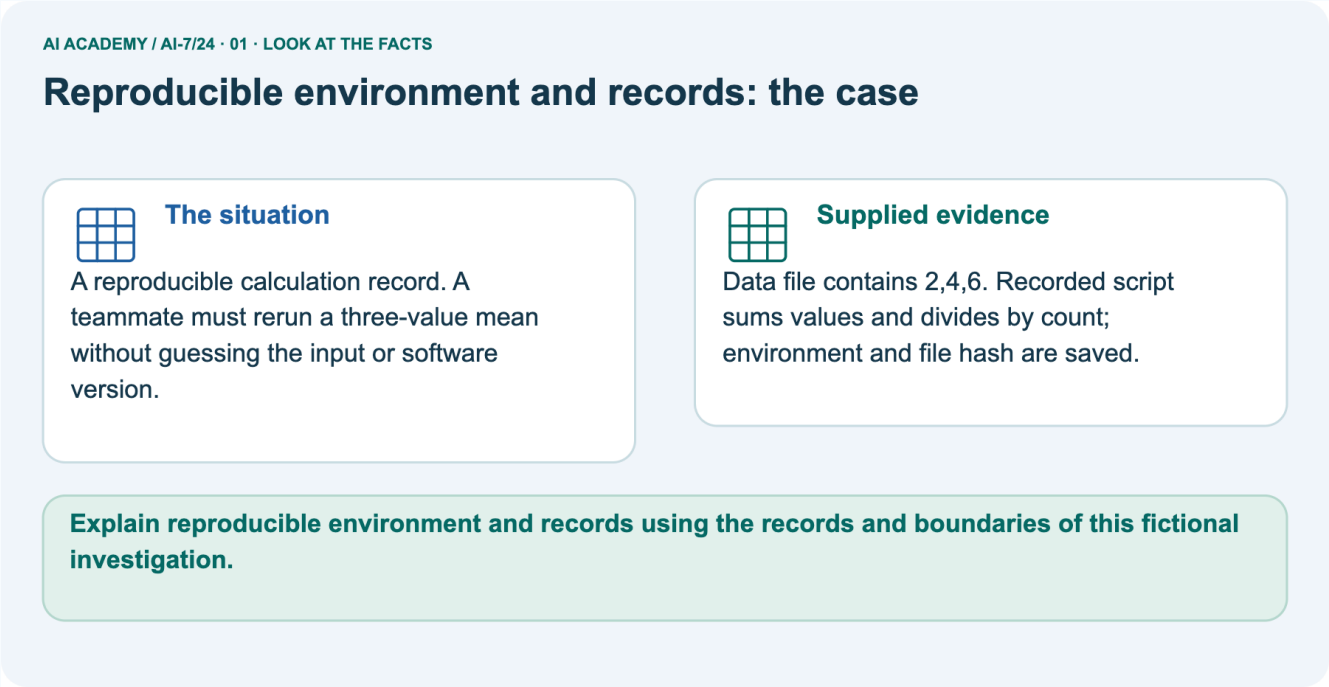
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Reproducible environment and records: the case



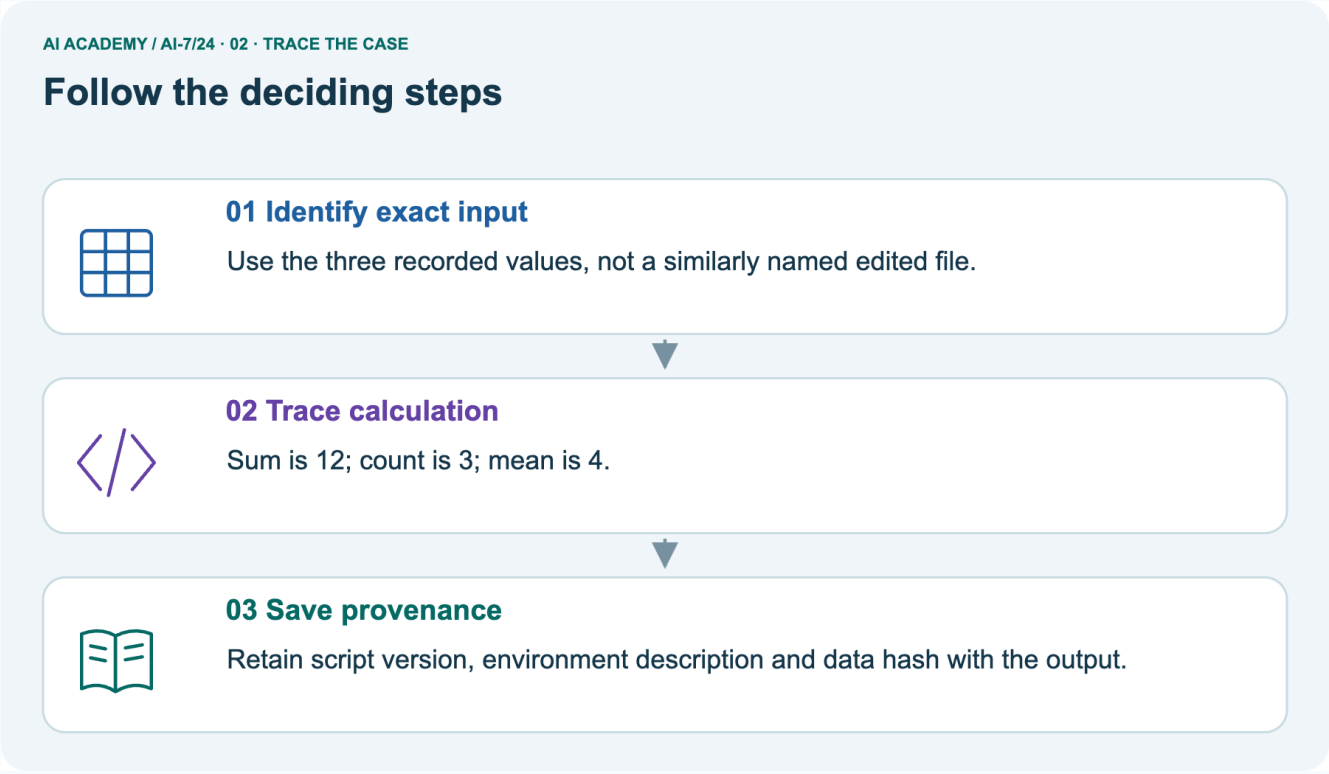
Follow the labelled picture with the matching explanation. Open the vector diagram

A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version. Supplied facts: Data file contains 2,4,6.

Recorded script sums values and divides by count; environment and file hash are saved.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

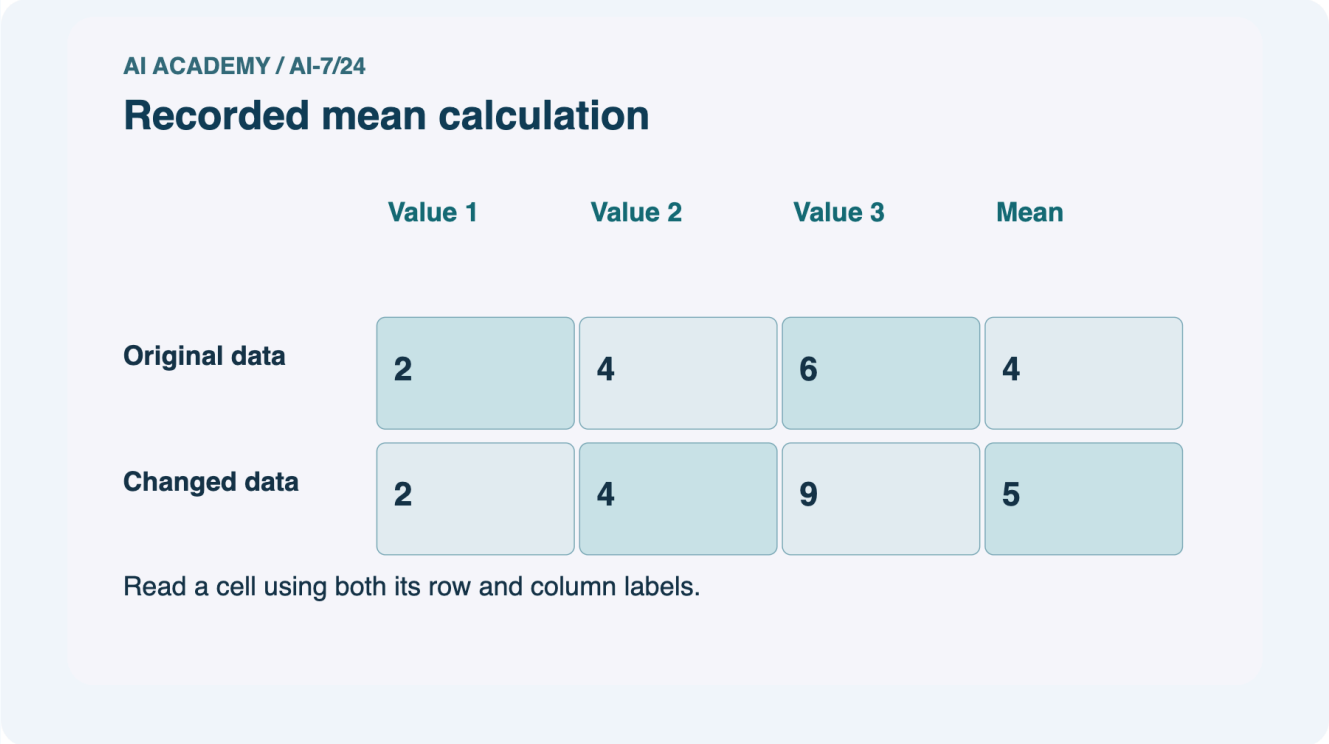


Follow the labelled picture with the matching explanation. Open the vector diagram

Identify exact **input**: Use the three recorded values, not a similarly named edited file. → Trace calculation: Sum is 12; count is 3; mean is 4. → Save provenance: Retain script version, environment description and data hash with the **output**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Recorded mean calculation

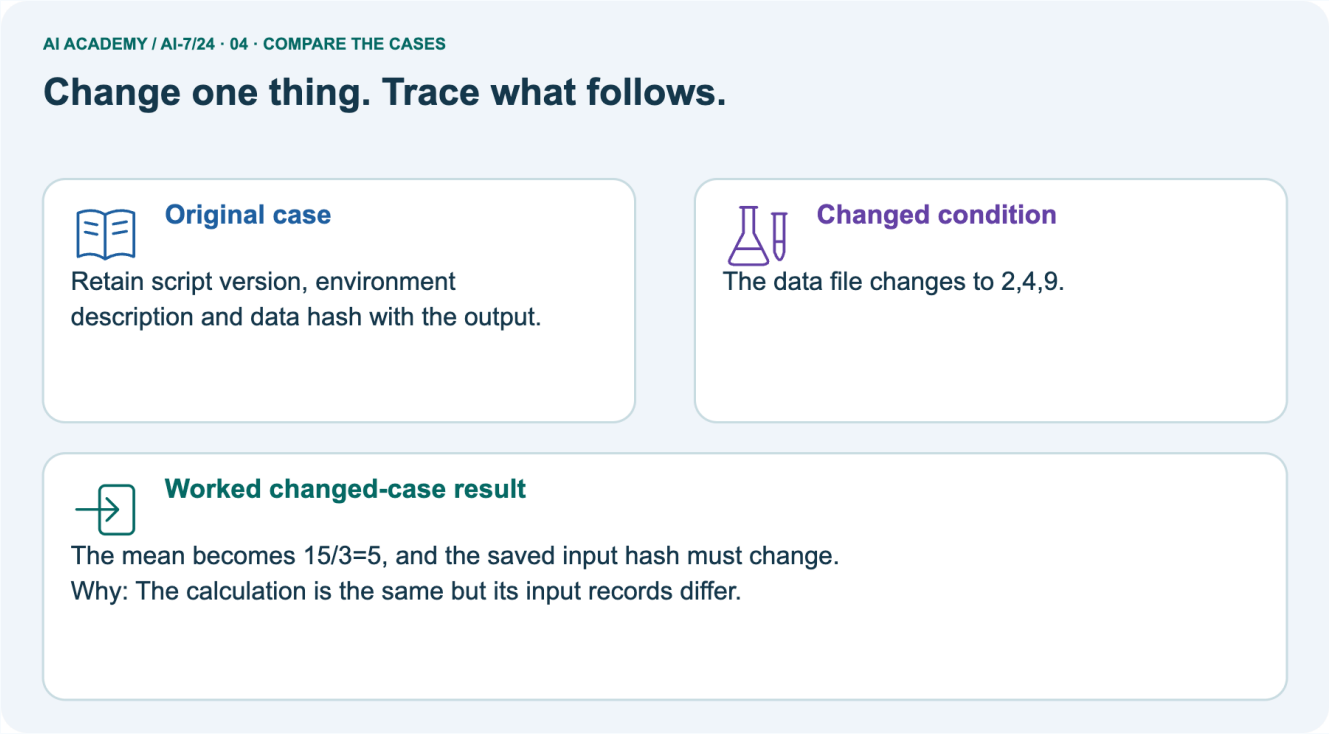


Follow the labelled picture with the matching explanation. Open the vector diagram

Recorded mean calculation. Original data: Value 1 = 2, Value 2 = 4, Value 3 = 6, Mean = 4. Changed data: Value 1 = 2, Value 2 = 4, Value 3 = 9, Mean = 5.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



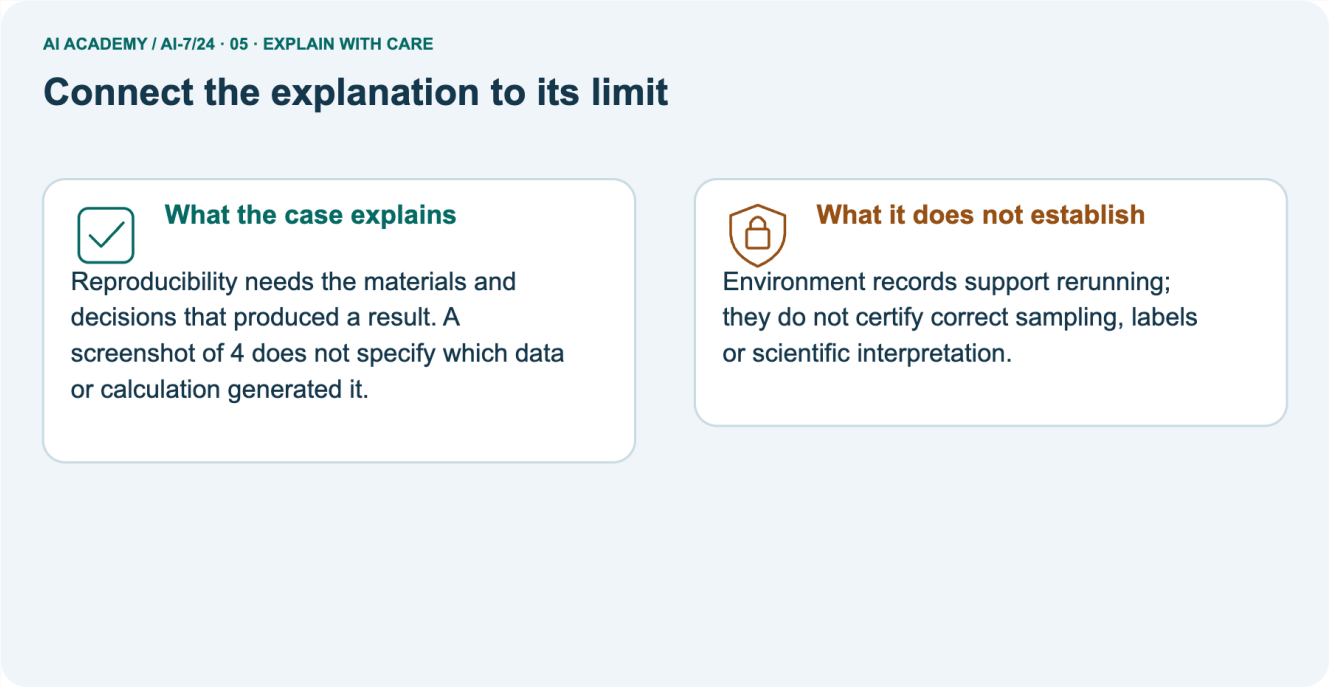
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retain script version, environment description and data hash with the **output**. Changed condition: The data file changes to 2,4,9. Expected result: The mean becomes 15/3=5, and the saved **input** hash must change.

Reason: The calculation is the same but its **input** records differ.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



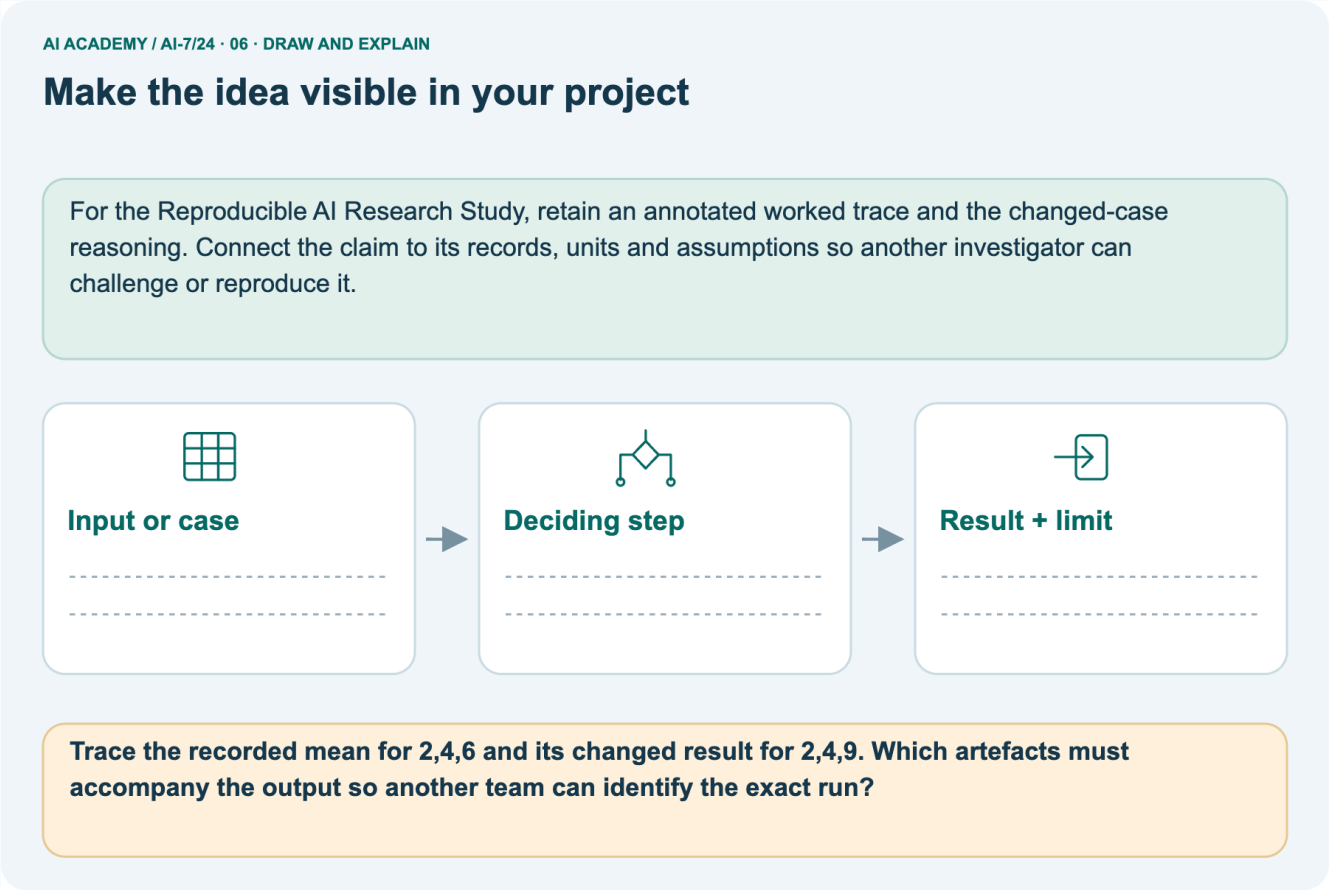
Follow the labelled picture with the matching explanation. Open the vector diagram

Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Limit: Environment records support rerunning; they do not certify correct sampling, labels or scientific interpretation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Reproducible environment and records: the case

02. Follow the deciding steps

03. Recorded mean calculation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.